

Tushar Singla

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EDUCATION

Chitkara University, Rajpura, Punjab
- Bachelor of Engineering (Computer Science Engineering)
Delhi Public School, Yamunanagar
- 12th
Delhi Public School, Yamunanagar
- 10th

Sep 2021 - Jul 2025
Overall CGPA: 9.51
Apr 2020 - Jul 2021
Percentage: 82%
Apr 2018 - May 2019
Percentage: 84%

Work Experience

AI & Software Engineer

Tata Chemicals (Aug 2025 – Present)

- Delivered AI-powered and full-stack applications to streamline operations and reduce manual effort.
- Constructed scalable data pipelines to transform raw inputs into actionable insights for business use.
- Leveraged machine learning techniques to enable predictive analytics and improve decision-making accuracy.
- Created interactive dashboards and reporting solutions to monitor system performance and user activity.
- Ensured secure and robust system design through access control mechanisms and cloud-based integrations.

SKILLS

- **Programming Languages:** Python, JavaScript, SQL
- **Frameworks & Libraries:** React, Angular, Nodejs, Fast Api, Flask
- **Data & AI/ML:** Machine Learning, Data Analysis, Data Processing, ETL Pipelines
- **Databases:** MongoDB Atlas on Azure, Azure SQL, Azure PostgreSQL, SQL Server
- **Cloud & Platforms:** Microsoft Azure, Azure OpenAI, Power BI, Power Apps, Power Automate
- **Web Technologies:** HTML, CSS, Bootstrap, Tailwind CSS
- **Core Concepts:** OOPS, DBMS, Operating System, Computer Networks
- **Soft Skills:** Problem-Solving, Collaboration, Communication

PROJECTS

Vibration Alerting & Predictive Maintenance System

Tech Stack: Python, PostgreSQL, Power BI, Machine Learning (XGBoost), Data Lakehouse, PAM Server

- Automated extraction and transformation of vibration data from PDF reports into structured database records.
- Established a secure deployment environment and integrated data storage with Lakehouse architecture.
- Enabled real-time monitoring and alerting through dashboards and automated email notifications.
- Applied XGBoost modeling to forecast equipment health categories and support preventive maintenance.

Crane Connect – Resource Booking System

Tech Stack: React 18, TypeScript, Fast Api, Azure PostgreSQL, Tailwind CSS, JWT, Docker

- Built an enterprise crane booking platform enabling teams to request, schedule, and track crane allocations across multiple departments.
- Implemented real-time calendar scheduling with hour-level booking visualization and conflict detection to prevent overlapping allocations.
- Developed secure backend services and RESTful APIs to manage crane requests, approvals, lifecycle tracking, and role-based access control.
- Designed admin dashboards with KPIs such as utilization, capacity matching, and on-time execution, improving operational visibility and planning.

Samarth Sarathi – Enterprise GenAI RAG Platform

Tech Stack: Angular 19, Fast Api, python, Azure OpenAI, Azure AI Search, Azure SQL, Power BI, Vector Database, Docker

- Architected a GenAI-driven RAG platform to facilitate intelligent retrieval of SOPs and equipment manuals.
- Devised a processing pipeline to convert scanned and non-scanned documents into searchable vector representations.
- Built a conversational interface delivering contextual responses with precise source references.
- Integrated authentication, authorization, and analytics modules to enhance security and track adoption.

CERTIFICATIONS

Coding Nijas — Introduction to Python	2021
Cisco Networking Academy – JavaScript Essentials	2022
Coursera - Java Programming: Solving Problems with Software	2023
Coursera - Google Digital Marketing and E-commerce	2023